$(X, \mathcal{O}_X)$ ringed sp. $\mathcal{F}, \mathcal{G} \in \mathcal{O}_X$-mod.

$\mathrm{Hom}(\mathcal{F}, -) : \mathcal{O}_X\text{-mod} \to \mathrm{Ab}$ left exact $\rightsquigarrow \mathrm{Ext}^i_{(\mathcal{F},-)} = R^i\,\mathrm{Hom}(\mathcal{F},-)$ i.e. $\mathrm{Ext}^i(\mathcal{F},\mathcal{G}) := h^i(\mathrm{Hom}(\mathcal{F}, I^\bullet))$ where $\mathcal{G} \to I^\bullet$. $\mathrm{Ext}^0(\mathcal{F},\mathcal{G}) = \mathrm{Hom}(\mathcal{F},\mathcal{G})$

$\mathcal{H}om(\mathcal{F}, -) : \mathcal{O}_X\text{-mod} \to \mathcal{O}_X\text{-mod}$. $\rightsquigarrow \mathcal{E}xt^i_{(\mathcal{F},-)} = R^i\,\mathcal{H}om(\mathcal{F},\mathcal{G})$. $\mathcal{E}xt^i(\mathcal{F},\mathcal{G}) := h^i(\mathcal{H}om(\mathcal{F}, I^\bullet))$ $\mathcal{G} \to I^\bullet$. $\mathcal{E}xt^0(\mathcal{F},\mathcal{G}) = \mathcal{H}om(\mathcal{F},\mathcal{G})$.

$U \mapsto \mathrm{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, -|_U)$.

$I \in \mathcal{O}_X$-mod inj $\Rightarrow$ $\mathrm{Ext}^i(\mathcal{F}, I) = 0$ $i>0$
$\mathcal{E}xt^i(\mathcal{F}, I) = 0$ $i>0$.

Lem. $I \in \mathcal{O}_X$-mod inj $\Rightarrow$ $I|_U \in \mathcal{O}_X|_U$-mod inj. **Proof.**

$$\begin{tikzcd} \mathcal{F} \arrow[r] \arrow[d] & \mathcal{G} \\ I|_U & \end{tikzcd} \quad \overset{j_!}{\rightsquigarrow} \quad \begin{tikzcd} j_!\mathcal{F} \arrow[r] \arrow[d] & j_!\mathcal{G} \\ j_!I|_U \arrow[d] & \\ I & \end{tikzcd} \quad \overset{|_U}{\rightsquigarrow} \quad \mathcal{G} \to I|_U.$$

Prop. $\mathcal{E}xt^i_X(\mathcal{F},\mathcal{G})|_U = \mathcal{E}xt^i_U(\mathcal{F}|_U, \mathcal{G}|_U)$.

Pf. $i=0$: $\mathcal{H}om_X(\mathcal{F},\mathcal{G})|_U = \mathcal{H}om_U(\mathcal{F}|_U, \mathcal{G}|_U)$
$\overset{\wr}{V} \mapsto \mathcal{H}om_V(\mathcal{F}, \mathcal{G}|_V)$ $\quad V \mapsto \mathcal{H}om_V(\mathcal{F}|_V, \mathcal{G}|_V)$.

etc $\underbrace{\mathcal{E}xt^i_X(\mathcal{F}, I)|_U}_{0\ \checkmark} = 0 = \mathcal{E}xt^i_U(\mathcal{F}|_U, \underbrace{I|_U}_{\text{inj}})$ ✓

Prop. $\mathcal{G} \in \mathcal{O}_X$-mod $\rightsquigarrow$ (1) $\mathcal{E}xt^0(\mathcal{O}_X, \mathcal{G}) = \mathcal{G}$ **Pf.** $\mathcal{H}om_X(\mathcal{O}_X, \mathcal{G})(U) = \mathrm{Hom}_{\mathcal{O}_X|_U}(\mathcal{O}_X|_U, \mathcal{G}|_U) = \mathcal{G}(U)$
(2) $\mathcal{E}xt^i(\mathcal{O}_X, \mathcal{G}) = 0$ $i>0$ **Pf.** $R^i\,\mathrm{Id} = 0$ $i>0$
(3) $\mathrm{Ext}^i(\mathcal{O}_X, \mathcal{G}) = H^i(X, \mathcal{G})$. **Pf.** $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{G}) = \Gamma(X, \mathcal{G})$ so both are $R^i\Gamma$.

Prop. $\mathrm{Ext}^i(-,\mathcal{G})$ / $\mathrm{Ext}^i(-,\mathcal{G})$ are δ-fctr **Pf.** functor: (contravariant) $\mathcal{F} \to \mathcal{F}'$ $\Longrightarrow$

$$\begin{tikzcd} \mathrm{Hom}(\mathcal{F}', I^\bullet) \arrow[d] \\ \mathrm{Hom}(\mathcal{F}, I^\bullet) \end{tikzcd} \Longrightarrow \begin{tikzcd} h^i(\mathrm{Hom}(\mathcal{F}', I^\bullet)) = \mathrm{Ext}^i(\mathcal{F}', \mathcal{G}) \arrow[d] \\ h^i(\mathrm{Hom}(\mathcal{F}, I^\bullet)) = \mathrm{Ext}^i(\mathcal{F}, \mathcal{G}) \end{tikzcd}$$

$\Rightarrow$

$$\begin{tikzcd} \mathcal{H}om(\mathcal{F}', I^\bullet) \arrow[d] \\ \mathcal{H}om(\mathcal{F}, I^\bullet) \end{tikzcd} \quad \text{over } U: \quad \begin{tikzcd} \mathrm{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}'|_U, I^\bullet|_U) \arrow[d] \\ \mathrm{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, I^\bullet|_U) \end{tikzcd} \quad \text{from } \mathcal{F}'|_U \to \mathcal{F}|_U. \ \checkmark$$

δ-fctr. $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$.

$0 \to \mathrm{Hom}(\mathcal{F}'', I^\bullet) \to \mathrm{Hom}(\mathcal{F}, I^\bullet) \to \mathrm{Hom}(\mathcal{F}', I^\bullet) \to 0$ ✓

Ext: ets $\mathcal{H}om(-, I^\bullet)$ exact. but exact on sections as $\mathrm{Hom}_{\mathcal{O}_X|_U}(-|_U, \underbrace{I|_U}_{\text{inj} \in \mathcal{O}_X|_U\text{-mod}})$.

Prop. loc free fr. $\mathcal{L}_\bullet \to \mathcal{F} \to 0$ $\rightsquigarrow$ $\mathcal{E}xt^i(\mathcal{F},\mathcal{G}) = h^i(\mathcal{H}om(\mathcal{L}_\bullet, \mathcal{G}))$.

Pf. $i=0$: $\mathcal{H}om(\mathcal{F},\mathcal{G}) = h^0(\mathcal{H}om(\mathcal{L}_\bullet, \mathcal{G}))$.

$0 \to \mathcal{H}om(\mathcal{F},\mathcal{G}) \to \mathcal{H}om(\mathcal{L}_0, \mathcal{G}) \to \mathcal{H}om(\mathcal{L}_1, \mathcal{G})$ ✓

$\mathrm{im} = \ker = h^0$.

coefficients in $\mathcal{G}$ as $\mathcal{H}om(-, I)$ exact.

Lem. $\mathcal{L}$ loc free fin rk. I inj $\rightsquigarrow$ $\mathcal{L} \otimes I$ inj **Pf.** $\mathrm{Hom}(-, \mathcal{L} \otimes I) \cong \mathrm{Hom}(- \otimes \mathcal{L}^\vee, I)$ exact.

Prop $\mathrm{Ext}^i(\mathcal{F} \otimes \mathcal{L}, \mathcal{G}) = \mathrm{Ext}^i(\mathcal{F}, \mathcal{L}^\vee \otimes \mathcal{G})$ **Pf.** $\mathrm{Hom}(\mathcal{F} \otimes \mathcal{L}, \mathcal{G}) = \mathrm{Hom}(\mathcal{F}, \mathcal{H}om(\mathcal{L}, \mathcal{G})) = \mathrm{Hom}(\mathcal{F}, \mathcal{L}^\vee \otimes \mathcal{G})$. both eff in $\mathcal{G}$ as $\mathcal{L}^\vee \otimes$ exact

$\mathcal{E}xt^i(\mathcal{F} \otimes \mathcal{L}, \mathcal{G}) = \mathcal{E}xt^i(\mathcal{F}, \mathcal{L}^\vee \otimes \mathcal{G}) = \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) \otimes \mathcal{L}^\vee$. **Pf.** $\mathcal{H}om(\mathcal{F} \otimes \mathcal{L}, \mathcal{G})(U) = \mathrm{Hom}(\mathcal{F}|_U \otimes \mathcal{L}|_U, \mathcal{G}|_U) = \mathrm{Hom}(\mathcal{F}|_U, \mathcal{L}^\vee|_U \otimes \mathcal{G}|_U) = \mathrm{Hom}(\mathcal{F}|_U, \mathcal{G}|_U) \otimes \mathcal{L}^\vee|_U$. eff in $\mathcal{G}$ ✓.

Prop. X noeth sch. $\mathcal{F}$ coh. $\mathcal{G} \in \mathcal{O}_X$-mod. $x \in X$. $\mathcal{E}xt^i(\mathcal{F},\mathcal{G})_x = \mathrm{Ext}^i_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{G}_x)$. **Pf.** wlog X affine

Serre duality

$B: V \times W \to k$ bilin nondegen $\rightsquigarrow$ $V \xrightarrow{\sim} W^\vee$ $\quad W \to V^\vee$
$v \mapsto (w \mapsto B(v,w))$ $\quad (w \mapsto (v \mapsto B(v,w)))$

Matsumura

C ring
N ideal $\ni N^2 = 0$. Then $N \in C$-mod $\rightsquigarrow N \in C/N$-mod by $(c+N)n = cn \in N$.
indeed if $c+N = c+n'+N$ then
$(c+N)n = cn$
$= cn + nn'$
$= (c+n'+N)n$.

Conversely, $N \in C'$-mod $\rightsquigarrow$ extension of C' by N is (C, ε, i) $N \to C$
ring $C \to C'$

$N \xrightarrow{i} C \xrightarrow{\varepsilon} C'$ s.t. $(\ker\varepsilon)^2 = 0$, so by above $\ker\varepsilon \in C/\ker\varepsilon \cong C'$-mod structure on $\ker\varepsilon$, and $i: N \xrightarrow{\sim} \ker\varepsilon$. as C'-mod.

i.e. SES of C'-mod
$0 \to N \xrightarrow{i} C \xrightarrow{\varepsilon} C' \to 0$.

equiv if

$$\begin{tikzcd} 0 \arrow[r] & N \arrow[r, "i"] \arrow[d, equal] & C_1 \arrow[r, "\varepsilon_1"] \arrow[d, "\wr"] & C' \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & N \arrow[r, "i_2"] & C_2 \arrow[r, "\varepsilon_2"] & C' \arrow[r] & 0 \end{tikzcd}$$

trivial extn

$N \to C' \oplus N \to C'$ abgp, give ring str $(a,x)(b,y) = (ab, ay+bx)$.
an abs is $\cong$ triv $\Longleftrightarrow$ $\exists$ section $\quad C \underset{\iota}{\overset{\varepsilon}{\rightleftarrows}} C'$

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Serre Duality

recall: $X = \mathbb{P}^r_A$, A noeth. $\exists$ perfect pairing of fg free A-mod

$$H^0(X, \mathcal{O}_X(n)) \times H^r(X, \mathcal{O}_X(-n-r-1)) \longrightarrow H^r(X, \mathcal{O}_X(-r-1)) \cong A.$$

- if $n<0$, trivial as all 0.
- $n \geq 0$: $H^0(X, \mathcal{O}_X(n)) = \operatorname{span}\{x_0^{m_0} \cdots x_r^{m_r} : m_i \geq 0, \sum m_i = n\}$

$$H^r(X, \mathcal{O}_X(-n-r-1)) = \operatorname{span}\{x_0^{l_0} \cdots x_r^{l_r} : l_i < 0, \sum l_i = -n-r-1\}$$

$$(x_0^{l_0} \cdots x_r^{l_r})(x_0^{m_0} \cdots x_r^{m_r}) = x_0^{l_0+m_0} \cdots x_r^{l_r+m_r} \quad (\deg -r-1).$$

Thm now let $X = \mathbb{P}^n_k$, $\omega_X = \Lambda^n \Omega_{X/k} \cong \mathcal{O}_X(-n-1)$.

(1) from computation of cohomology of $\mathbb{P}^n$, $H^n(X, \omega_X) \cong k$.

(2) $\forall \mathcal{F} \in \operatorname{Coh} X$, $\exists$ perfect pairing

$$\operatorname{Hom}(\mathcal{F}, \omega_X) \times H^n(X, \mathcal{F}) \longrightarrow H^n(X, \omega) \cong k$$

(3) $\forall i \geq 0$ $\exists$ natural iso

$$\operatorname{Ext}^i(\mathcal{F}, \omega_X) \xrightarrow{\sim} H^{n-i}(X, \mathcal{F})^\vee.$$

Proof. (2) $\varphi \in \operatorname{Hom}(\mathcal{F}, \omega_X) \rightsquigarrow H^n(X, \mathcal{F}) \longrightarrow H^n(X, \omega_X) \cong k$

if $\mathcal{F} = \mathcal{O}(q)$ then $\operatorname{Hom}(\mathcal{F}, \omega_X) \cong \operatorname{Hom}(\mathcal{O}, \omega_X(-q)) \cong H^0(X, \omega(-q))$ so this is the known pairing.

$\operatorname{Hom}(\mathcal{E} \otimes \mathcal{F}, \mathcal{G}) \cong \operatorname{Hom}(\mathcal{F}, \mathcal{H}om(\mathcal{E}, \mathcal{G}))$, $\mathcal{E}$ loc. free

for $\mathcal{F} = \bigoplus_i \mathcal{O}(q_i)$, $\operatorname{Hom}(\mathcal{F}, \omega_X) \cong \bigoplus_i \operatorname{Hom}(\mathcal{O}(q_i), \omega_X)$ so we know the pairing.

in general $\mathcal{F} \in \operatorname{Coh} X$ so $\mathcal{E}_1 \to \mathcal{E}_0 \to \mathcal{F} \to 0$, where $\mathcal{E}_i$ is the prev. form. so

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{Hom}(\mathcal{F}, \omega) \arrow[r] \arrow[d] & \operatorname{Hom}(\mathcal{E}_0, \omega) \arrow[r] \arrow[d, "\wr"] & \operatorname{Hom}(\mathcal{E}_1, \omega) \arrow[d, "\wr"] \\
0 \arrow[r] & H^n(X, \mathcal{F})^\vee \arrow[r] & H^n(X, \mathcal{E}_0)^\vee \arrow[r] & H^n(X, \mathcal{E}_1)^\vee
\end{tikzcd}$$

iso by 5-lem. $\square$

Lem $X \hookrightarrow \mathbb{P}^N_k$ codim r. Then $\mathcal{E}xt^i_{\mathbb{P}^N}(\mathcal{O}_X, \omega_{\mathbb{P}^N}) = 0$ $i<r$.

Proof. $\mathcal{F}^i := \mathcal{E}xt^i_{\mathbb{P}^N}(\mathcal{O}_X, \omega_{\mathbb{P}^N})$ coherent as $\mathcal{O}_X$, $\omega_{\mathbb{P}^N}$ coherent,

so $\mathcal{F}^i(n) \cong \mathcal{E}xt^i_{\mathbb{P}^N}(\mathcal{O}_X, \omega_{\mathbb{P}^N}(n))$ glob. gen for $n \gg 0$

$\Rightarrow$ ets $P(\mathcal{E}xt^i_{\mathbb{P}^N}(\mathcal{O}_X, \omega_{\mathbb{P}^N}(n))) = 0$.

but this is $\operatorname{Ext}^i_{\mathbb{P}^N}(\mathcal{O}_X, \omega_{\mathbb{P}^N}(n))$ for large n, which is

$\cong$ by Serre duality on $\mathbb{P}^N$

$H^{N-i}(\mathbb{P}^N, \mathcal{O}_X(-n)) = 0$ for $i<r$ $\rightsquigarrow$ $N-i > \dim X$. $\square$

by Grothendieck van.

Lem. $X \hookrightarrow \mathbb{P}^N_k$ codim r. $\omega_X^\bullet = \mathcal{E}xt^r_{\mathbb{P}^N}(\mathcal{O}_X, \omega_{\mathbb{P}^N})$ then

$$\operatorname{Hom}_X(\mathcal{F}, \omega_X^\bullet) \cong \operatorname{Ext}^r_{\mathbb{P}^N}(\mathcal{F}, \omega_{\mathbb{P}^N}) \quad \forall \mathcal{F} \in \mathcal{O}_X\text{-mod}.$$

Proof. $0 \to \omega_{\mathbb{P}^N} \to I^\bullet$ then $\operatorname{Ext}^r_{\mathbb{P}^N}(\mathcal{F}, \omega_{\mathbb{P}^N}) = h^r(\operatorname{Hom}_{\mathbb{P}^N}(\mathcal{F}, I^\bullet))$

$$= h^r(\operatorname{Hom}_X(\mathcal{F}, \mathcal{H}om_{\mathbb{P}^N}(\mathcal{O}_X, I^\bullet)))$$

Smoothness, flatness

$X \xrightarrow{f} Y$ smooth of rel dim d at x if $\exists$

$$\begin{tikzcd}
U \arrow[r, "f"] \arrow[d, "\in"] & V \arrow[d, "\in"] \\
\operatorname{Spec} S \arrow[r] & \operatorname{Spec} R \\
x \arrow[r, mapsto] & f(x)
\end{tikzcd}$$

$\operatorname{Spec} R[T_1 \cdots T_n]/(f_1 \cdots f_{n-d})$

s.t. $\left(\frac{\partial f_i}{\partial T_j}(x)\right)_{i,j} \in M_{(n-d)\times n}(\kappa(x))$ has rk $n-d$.

ex. $X \to \operatorname{Spec} k$ flat as $\forall x \in X$, $\mathcal{F}_x \in k$-mod is free

ex. $\mathbb{A}^n_Y \to Y$ flat as locally $\operatorname{Spec}(A[T_1 \cdots T_n]) \to \operatorname{Spec} A$, flat.

$\cong A^n$

$B \xrightarrow{\phi} A$. $M \in A$-mod is flat$/B \Leftrightarrow \tilde{M}$ flat$/\operatorname{Spec} B$, i.e. $\forall \mathfrak{p} \in \operatorname{Spec} A$, $\tilde{M}_\mathfrak{p} = M_\mathfrak{p}$ is flat $B_{\phi^{-1}(\mathfrak{p})}$-mod.

$\Rightarrow M$ flat$/B$. $\rightsquigarrow \mathfrak{q} = \phi^{-1}(\mathfrak{p}) \rightsquigarrow \phi(B-\mathfrak{q}) \subset A - \mathfrak{p}$.

$M \otimes_B B_\mathfrak{q}$ flat$/B_\mathfrak{q}$.

$\| $

$M_\mathfrak{q}$

$M_\mathfrak{p}$ is a localization of $M_\mathfrak{q}$, so flat $B_\mathfrak{q}$-mod ✓

$\Leftarrow$: M not B-flat, then $\exists$ B-mod: $0 \to N' \to N$ st $\ker(M \otimes_B N' \to M \otimes_B N) =: K \neq 0$.

viewed as A-mod, $\exists$ $\mathfrak{p} \in A$ st. $K_\mathfrak{p} \neq 0$.

$\rightsquigarrow 0 \to K_\mathfrak{p} \to M_\mathfrak{p} \otimes_B N' \to M_\mathfrak{p} \otimes_B N$

$\rightsquigarrow 0 \to K_\mathfrak{p} \to M_\mathfrak{p} \otimes_{B_\mathfrak{q}} N'_\mathfrak{q} \to M_\mathfrak{p} \otimes_{B_\mathfrak{q}} N_\mathfrak{q}$

$\| $

$\Rightarrow 0 \quad \mathcal{E}.$

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- $M \in A$-mod flat $\Leftrightarrow$ $\forall f.g$ $\mathfrak{a} \subset A$, $\mathfrak{a} \otimes M \hookrightarrow M$.
- $M \in A$-mod flat, $A \to B$ $\Rightarrow$ $M \otimes_A B$ flat B-mod.
- $A \to B$ flat, N flat B-mod $\Rightarrow$ N flat A-mod.
- M flat A-mod $\Rightarrow$ M_p flat A_p-mod $\forall p \in \operatorname{Spec} A$.
- $0 \to M' \to M \to M'' \to 0$. M', M'' flat $\Rightarrow$ M flat; M, M'' flat $\Rightarrow$ M flat
- M f.g $/A$ loc noeth. M flat $\Leftrightarrow$ free.
- $A \to S^{-1}A$ flat
- $A \to B$, $M \in B$-mod flat$/A$, $S \subset B$ then $S^{-1}M$ flat$/A$.
- $\hat{A}$ (a-adic completion of noeth ring) flat$/A$.
- M flat$/A$ PID $\Leftrightarrow$ torsion free
- $\hookrightarrow$ flat (iso on stalks)
- $\begin{tikzcd} X' \arrow[r, "p_1"] \arrow[d, "p_2"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"] & Y \end{tikzcd}$, $\mathcal{F} \in \mathcal{O}_X$-mod flat$/Y$. $\Rightarrow$ $p_1^* \mathcal{F}$ flat$/Y'$.
- $X \to Y \xrightarrow{\text{flat}} Z$. $\mathcal{F} \in \mathcal{O}_X$-mod flat$/Y$ $\Rightarrow$ flat$/Z$.
- $A \to B$, $M \in B$-mod flat$/A$ $\Leftrightarrow$ $\tilde{M}$ flat$/\operatorname{Spec} A$.
- X noetherian. $\mathcal{F} \in \operatorname{Coh} X$ flat $\Leftrightarrow$ loc free.

$f: X \to Y$. $\mathcal{F} \in \operatorname{Ab}(X)$ s.t. $R^i f_* \mathcal{F} = 0$ $i > 0$. then $H^i(X, \mathcal{F}) \cong H^i(Y, f_* \mathcal{F})$.

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{F} \arrow[r] \arrow[d] & I^\bullet \arrow[d] \\ & f_*\mathcal{F} \arrow[r] & f_* I^\bullet \end{tikzcd}$$

$\rightsquigarrow$ $0 \to f_*\mathcal{F} \to f_* I^\bullet$ exact as its cohomology is $R^i f_* \mathcal{F} = 0$.

$\Big\downarrow \Gamma$

$$\begin{tikzcd} 0 \arrow[r] & \Gamma(X, \mathcal{F}) \arrow[r] \arrow[d, equal] & \Gamma(X, I^\bullet) \arrow[d, equal] \\ 0 \arrow[r] & \Gamma(Y, f_*\mathcal{F}) \arrow[r] & \Gamma(X, I^\bullet) \end{tikzcd} \xrightarrow{H^i} \begin{array}{c} H^i(X, \mathcal{F}) \\ \| \\ H^i(Y, f_*\mathcal{F}). \end{array}$$

ex. f affine, $\mathcal{F} \in \operatorname{QCoh} X$, X noeth. $R^i f_* \mathcal{F} = (U \mapsto H^i(f^{-1}(U), \mathcal{F}|_{f^{-1}(U)}))^\dagger$

U aff $\Rightarrow$ $f^{-1}(U)$ aff, so $= 0$ by Serre vanishing.

$f: X \to Y$, $\mathcal{F} \in \mathcal{O}_X$-mod, $\mathcal{E}$ loc free $\mathcal{O}_Y$-mod of fin. rk.

$R^i f_*(\mathcal{F} \otimes f^*\mathcal{E}) \cong R^i f_*(\mathcal{F}) \otimes \mathcal{E}$.

i.e., $f_*(\mathcal{F} \otimes f^*\mathcal{E}) \cong f_*\mathcal{F} \otimes \mathcal{E}$ proj form.

$0 \to \mathcal{F} \to I^\bullet \rightsquigarrow$ LHS is $H^i(f_*(I^\bullet \otimes f^*\mathcal{E})) = H^i(f_* I^\bullet \otimes \mathcal{E}) = H^i(f_* I^\bullet) \otimes \mathcal{E}$ as $\mathcal{E}$ loc free.
$= R^i f_*(\mathcal{F})$.

$\begin{tikzcd} X \arrow[d, "\text{projective}"] \\ \operatorname{Spec} k \end{tikzcd}$ integral $\dim \geq 1$. $\mathcal{L} \in \operatorname{Pic} X$ ample. Then $H^0(X, \mathcal{L}^{-1}) = 0$.

if not, $s \in \Gamma(X, \mathcal{L}^{-1}) \rightsquigarrow \mathcal{O}_X \to \mathcal{L}^{-1}$

$a \mapsto as$

Prop. $\begin{tikzcd} X' \arrow[r, "v"] \arrow[d, "g"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "u", "\text{flat}"'] & Y \end{tikzcd}$ · sep · f.t.

X, Y noeth. $\mathcal{F} \in \operatorname{QCoh} X$. Then $u^* R^i f_* \mathcal{F} \cong R^i g_* v^* \mathcal{F}$.

Proof. Wlog $Y = \operatorname{Spec} A$, $Y' = \operatorname{Spec} A'$. $u \to A \to A'$ flat. LHS $= u^* H^i(X, \mathcal{F})^\sim = (H^i(X, \mathcal{F}) \otimes_A A')^\sim$

RHS $= H^i(X', v^*\mathcal{F})^\sim$

so ETS $H^i(X, \mathcal{F}) \otimes_A A' \cong H^i(X', v^*\mathcal{F})$.

$H^i(C^\bullet(\mathfrak{U}, \mathcal{F})) \otimes_A A'$ $\quad$ $H^i(C^\bullet(v^*\mathfrak{U}, v^*\mathcal{F}))$

where $\mathfrak{U}$ aff cover. $\|$ $\quad$ $\|$

$H^i(C^\bullet(\mathfrak{U}, \mathcal{F}) \otimes_A A')$ as $C^p(v^*\mathfrak{U}, v^*\mathcal{F}) = \prod_{|J| = p+1} v^*\mathcal{F}($

A' flat$/A$

A- ring. $\mathbb{P}^n_A = \operatorname{Proj} A[x_0 \cdots x_n]$, $\deg(x_i) = 1$

$S(m) := A[x_0 \cdots x_n]$ $\leadsto$ grading $S(m)_d = S_{m+d}$.

$\mathcal{O}(m) := S(m)^\sim$

Now $\Gamma(\mathbb{P}^n_A, \mathcal{O}(m)) = S(m)_0 = S_m$

in particular $\Gamma(\mathbb{P}^n_A, \mathcal{O}(1))$ gen. by $x_0 \cdots x_n$.

in fact $\mathcal{O}(1)$ is gen by $x_0 \cdots x_n$, i.e. $\forall p \in \mathbb{P}^n_A$, $\mathcal{O}(1)_p$ is gen by $x_0 \cdots x_n$ over $\mathcal{O}_p$.

Now $\begin{tikzcd} X \arrow[r, "\varphi"] \arrow[dr] & \mathbb{P}^n_A \arrow[d] \\ & A \end{tikzcd}$ $\rightsquigarrow$ $\mathcal{L} := \varphi^* \mathcal{O}(1)$ invertible and $s_i := \varphi^*(x_i) \in \Gamma(X, \mathcal{L})$ generate $\mathcal{L}$

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25 Derivations

A ring
$M \in A$-mod

$D: A \longrightarrow M$ is derivation if
- $D(a+b) = D(a) + D(b)$
- $D(ab) = aD(b) + bD(a)$.

$\mathrm{Der}(A, M) = \{ D: A \to M \text{ derivations} \}$ is A-mod by
$(a \cdot D)(b) = a \cdot D(b)$.

More generally, $A \in k\text{-Mg}$, $k \xrightarrow{f} A$.

$D: A \longrightarrow M$ is k-derivation if
- $D(a+b) = D(a) + D(b)$
- $D(x \cdot a) = x \cdot D(a)$
- $D(ab) = aD(b) + bD(a)$.

$\longleftrightarrow$ $D \circ f = 0$. indeed $\Rightarrow$: $\forall x \in k$, $D \circ f(x) = D(x \cdot 1) = x \cdot D(1)$
cts $D(1) = 0$.
$= D(1 \cdot 1) = D(1) + D(1) \Rightarrow D(1) = 0$ provided char $\neq 2$.

$\Leftarrow$: $D \circ f = 0 \Rightarrow D(x \cdot a) = D(f(x) a)$
$= a D(f(x)) + f(x) D(a)$
$= x \cdot D(a)$.

$\mathrm{Der}_k(A, M) = \{ D: A \to M \ k\text{-der} \}$ is A-submod of $\mathrm{Der}(A, M)$.
$\overset{\shortparallel}{\mathrm{Der}_{\mathbb{Z}}(A, M)}$.

Special case
$M = A$: $\mathrm{Der}_k(A) := \mathrm{Der}_k(A, A)$
$D, D' \in \mathrm{Der}_k(A) \rightsquigarrow [D, D'] = DD' - D'D \in \mathrm{Der}_k(A)$ so $\mathrm{Der}_k(A)$ is Lie alg.
in general $D(a^n) = n a^{n-1} D_a$ $\quad$ A char $p \Rightarrow D(a^p) = 0$

$D^n(ab) = \sum_{i=0}^{n} \binom{n}{i} D^i(a) \cdot D^{n-i}(b)$. $\quad$ A char $p \Rightarrow D^p(ab) = bD^p(a) + a \cdot D^p(b)$ so $D^p \in \mathrm{Der}\, A$.

$A\text{-mod} \longrightarrow A\text{-mod}$
$M \longmapsto \mathrm{Der}_k(A, M)$ is representable. i.e. $\exists M_0 \in A$-mod, $d \in \mathrm{Der}_k(A, M_0)$, s.t.
$\forall M \in A$-mod, $D \in \mathrm{Der}_k(A, M)$, $\exists!\, f \in \mathrm{Hom}_A(M_0, M)$ s.t. $D = f \circ d$.

$$\begin{tikzcd} A \arrow[r, "D"] \arrow[d, "d"'] & M \\ M_0 \arrow[ur, dashed, "\exists!"'] & \end{tikzcd}$$

i.e.

$$\begin{array}{ccc} M & \longmapsto & \mathrm{Der}_k(A, M) \\ & \Updownarrow & \\ M & \longmapsto & \mathrm{Hom}_A(M_0, M) \end{array}$$

$\mu: A \otimes_k A \longrightarrow k$ $\quad$ k-alg hom. $\quad$ $I = \ker \mu$, $\Omega_{A/k} = I/I^2$, $B = A \otimes_k A / I^2$
$x \otimes y \longmapsto xy$.

$$\begin{tikzcd} A \arrow[r] \arrow[d] & M \\ \Omega_{A/k} \arrow[ur, dashed] & \end{tikzcd}$$

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$A \xrightarrow{f} B$ faithfully flat Then Amitsur (f) exact.

$$0 \longrightarrow A \xrightarrow{f} B \underset{b \mapsto 1\otimes b}{\overset{b \mapsto b\otimes 1}{\rightrightarrows}} B \otimes_A B \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; B \otimes_A B \otimes_A B \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \cdots$$

recall Amitsur(f) is obtained by taking alternating sums of the arrows.

i.e. $d^i : B^{\otimes n+1} \longrightarrow B^{\otimes n+2}$

$$x_0 \otimes \cdots \otimes x_n \longmapsto x_0 \otimes \cdots \otimes x_{i-1} \otimes 1 \otimes x_i \otimes \cdots \otimes x_n$$

then $\delta^n : B^{\otimes n+1} \longrightarrow B^{\otimes n+2}$ is $\sum_{i=0}^{n+1} (-1)^i d^i$.

that is, $0 \to A \underset{s}{\overset{f}{\rightleftarrows}} B \xrightarrow{\delta^0} B^{\otimes 2} \xrightarrow{\delta^1} \cdots$ is exact.

1. if $\exists\, s : B \to A$ s.t. $s \circ f = id_A$, then define null htpy $h^n : B^{\otimes n+1} \to B^{\otimes n}$

$$x_0 \longmapsto s(x_0) \qquad n = 0$$

$$x_0 \otimes \cdots \otimes x_n \longmapsto f \circ s(x_0)\, x_1 \otimes \cdots \otimes x_n \qquad n > 0.$$

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r, "f"] \arrow[d, equal] & B \arrow[r, "\delta^0"] \arrow[d, equal] \arrow[dl, "h^0"'] & B^{\otimes 2} \arrow[r, "\delta^1"] \arrow[d, equal] \arrow[dl, "h^1"'] & \cdots \\
0 \arrow[r] & A \arrow[r] & B \arrow[r] & B^{\otimes 2} \arrow[r, "\delta^1"] & {}
\end{tikzcd}$$

$$f \circ h^0(x_0) + h^1 \circ \delta^0(x_0) = f \circ s(x_0) + h^1(1 \otimes x_0 - x_0 \otimes 1)$$

$$= f \circ s(x_0) + x_0 - f \circ s(x_0)$$

$$= id_B(x_0)$$

$$\left(h^{n+1} \circ \delta^n + \delta^{n-1} \circ h^n\right)(x_0 \otimes \cdots \otimes x_n) = \sum_{i=0}^{n+1} h^{n+1}(x_0 \otimes \cdots \otimes x_{i-1} \otimes 1 \otimes x_i \otimes \cdots \otimes x_n) + \delta^{n-1}\left(f \circ s(x_0) \cdot x_1 \otimes \cdots \otimes x_n\right)$$

$$= x_0 \otimes \cdots \otimes x_n + \sum_{i=1}^{n+1} (-1)^i f \circ s(x_0)\, x_1 \otimes \cdots \otimes x_{i-1} \otimes 1 \otimes x_i \otimes \cdots \otimes x_n + \sum_{i=0}^{n-1} (-1)^i f \circ s(x_0)\, x_1 \otimes \cdots \otimes x_i \otimes 1 \otimes x_{i+1} \otimes \cdots \otimes x_n$$

$$= x_0 \otimes \cdots \otimes x_n .$$

2. in general. faith. flat mod reflects exactness, so cts $(A \to B) \otimes_A B$ exact. true as $\begin{array}{l} B \longrightarrow B \otimes_A B \\ b \longmapsto 1 \otimes b \end{array}$ has retr. $x \otimes y \longmapsto xy$.

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Hartshorne III.9.7.

$$\begin{tikzcd} Y \arrow[r, hook] & X \arrow[d, "f.t."] \\ & \operatorname{Spec} k \end{tikzcd}$$

$D = k[t]/t^2$. Def. infinitesimal defn of $Y \hookrightarrow X$ is

$$\begin{tikzcd} Y' \arrow[r, hook] \arrow[d, "\text{flat}"] & X \times_k D \\ D & \end{tikzcd}$$

, closed fibre is Y. $\hookrightarrow Y'_k \cong Y$

Show they're classified by $H^0(Y, \mathcal{N}_{Y/X})$

$$\| $$

$$\operatorname{Hom}_{\mathcal{O}_Y}(\mathcal{I}_Y/\mathcal{I}_Y^2, \mathcal{O}_Y).$$

Hartshorne III.9.8

A f.g. k-alg. write $A \cong P/J$, $P = k[t_1 \dots t_r]$ $\rightsquigarrow$ $0 \to J \to P \to A \to 0$.

$\hookrightarrow k \to A$ or $\operatorname{Spec}(A) \xrightarrow{\text{finite type}} \operatorname{Spec} k$.

2nd exact seq.

$$J/J^2 \xrightarrow{\delta} \Omega_{P/k} \otimes_P A \to \Omega_{A/k} \to 0$$

$$\bar{x} \longmapsto dx \otimes 1$$

$$\operatorname{Hom}_A(\Omega_{P/k} \otimes_P A, A) \to \operatorname{Hom}_A(J/J^2, A) \to T^1(A) \to 0$$

$\hookrightarrow$ Show this classifies defns, i.e. A' flat$/D$, $A' \otimes_D k \cong A$.

$$\begin{tikzcd} 0 \arrow[d] & 0 \arrow[d] \\ J \arrow[d] & tA' \arrow[d] \\ P \arrow[r] \arrow[d] & A' \arrow[d] \\ A \arrow[r] \arrow[d] & D \arrow[d] \\ 0 & 0 \end{tikzcd}$$

Bosch 7.6.1.

$\mathbb{A}^1_k \cup \mathbb{A}^1_k$ $\rightsquigarrow$ X. $H^q(\mathcal{U}, \mathcal{O}_X)$.

$\hookrightarrow k[x, x^{-1}] \to k[x, x^{-1}]$

$x \longmapsto x$

$$k[x] \oplus k[y] \longrightarrow k[x, x^{-1}] \longrightarrow 0$$

$$(x, y) \longmapsto y - x$$

$H^0(\mathcal{U}, \mathcal{O}_X) \cong k[x]$.

$H^1(\mathcal{U}, \mathcal{O}_X) \cong k[x, x^{-1}] / k[x] \oplus k[y] = k[x^{-1}]$.

$H^q(\mathcal{U}, \mathcal{O}_X) = 0 \quad q \geq 2$.

$x \longmapsto x^{-1}$

$$k[x] \oplus k[y] \longrightarrow k[x, x^{-1}] \longrightarrow 0$$

$$(x, y) \longmapsto y^{-1} - x$$

$H^0(\mathcal{U}, \mathcal{O}_X) = K$

$H^1(\mathcal{U}, \mathcal{O}_X) = 0$

$H^q(\mathcal{U}, \mathcal{O}_X) = 0 \quad q \geq 2$.

$\mathcal{U} \overset{j}{\hookrightarrow} X$, $Z = X \setminus \mathcal{U} \overset{i}{\hookrightarrow} X$.

$$0 \to j_! \mathcal{F}|_{\mathcal{U}} \to \mathcal{F} \to i_* \mathcal{F}|_Z \to 0.$$

$$0 \to \left(\begin{cases} \mathcal{F}(V) & V \subset \mathcal{U} \\ 0 & \text{else} \end{cases} \right)^+ \to \mathcal{F}(V) \to \mathcal{F}(Z \cap V) \to 0$$

$X = \mathbb{P}^1$. $\mathcal{K}$ const sheaf $K(X) = \mathcal{O}_{X,\eta}$.

$$\exists\ 0 \to \mathcal{O} \to \mathcal{K} \to \mathcal{K}/\mathcal{O} \to 0$$

$$\mathcal{K}/\mathcal{O} \cong \sum_{p \in X} i_p(K/\mathcal{O}_p)$$

$\mathbb{P}^1$ integral so every $\emptyset \neq \mathcal{U} \subset X$ contains η, so $\mathcal{O}(\mathcal{U}) \to \mathcal{O}_{X,\eta} = \mathcal{K}(\mathcal{U})$.

check inj on stalks: $\mathcal{O}_{X,p} \hookrightarrow \mathcal{O}_{X,\eta} = \operatorname{Frac}(\mathcal{O}_{X,p})$.

now $\mathcal{K}$ flasque as const on irred.

$\mathcal{K}(\mathcal{U}) = K \to K/\mathcal{O}_{X,p}$ $\rightsquigarrow$ $\mathcal{K} \to \sum_{p \in X} (i_p)_* (K/\mathcal{O}_{X,p})$. check $K/\mathcal{O} \cong \underbrace{\sum_{p \in X} (i_p)_* (K/\mathcal{O}_{X,p})}_{\text{obviously flasque.}}$ on stalks.

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